System Verification and Validation Plan for FPVR

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

[The intention of the VnV plan is to increase confidence in the software. However, this does not mean listing every verification and validation technique that has ever been devised. The VnV plan should also be a **feasible** plan. Execution of the plan should be possible with the time and team available. If the full plan cannot be completed during the time available, it can either be modified to "fake it", or a better solution is to add a section describing what work has been completed and what work is still planned for the future. —SS]

[The VnV plan is typically started after the requirements stage, but before the design stage. This means that the sections related to unit testing cannot initially be completed. The sections will be filled in after the design stage is complete. the final version of the VnV plan should have all sections filled in. —SS]

Contents

1 Symbols, Abbreviations, and Acronyms **iv**

2 General Information **1**
- 2.1 Summary . . . 1
- 2.2 Objectives . . . 1
- 2.3 Custom Documents . . . 2
- 2.4 Relevant Documentation . . . 2

3 Plan **3**
- 3.1 Verification and Validation Team . . . 3
- 3.2 SRS Verification . . . 3
- 3.3 Design Verification . . . 4
- 3.4 Verification and Validation Plan Verification . . . 4
- 3.5 Implementation Verification . . . 4
- 3.6 Automated Testing and Verification Tools . . . 4
- 3.7 Software Validation . . . 5

4 System Tests **5**
- 4.1 Tests for Functional Requirements . . . 5
  - 4.1.1 Area of Testing1 . . . 6
  - 4.1.2 Area of Testing2 . . . 6
- 4.2 Tests for Nonfunctional Requirements . . . 7
  - 4.2.1 Area of Testing1 . . . 7
  - 4.2.2 Area of Testing2 . . . 8
- 4.3 Traceability Between Test Cases and Requirements . . . 8

5 Unit Test Description **8**
- 5.1 Unit Testing Scope . . . 8
- 5.2 Tests for Functional Requirements . . . 9
  - 5.2.1 Module 1 . . . 9
  - 5.2.2 Module 2 . . . 10
- 5.3 Tests for Nonfunctional Requirements . . . 10
  - 5.3.1 Module ? . . . 10
  - 5.3.2 Module ? . . . 10
- 5.4 Traceability Between Test Cases and Modules . . . 11

6 Appendix **12**
6.1 Symbolic Parameters 12
6.2 Usability Survey Questions? 12

7 Appendix — Generative AI Use Disclosure **13**
7.1 AI Tools Used 13
7.2 How and Where Used 13
7.3 Verification of AI-Generated Content 13

8 Appendix — Reflection **14**

List of Tables

[Remove this section if it isn't needed —SS]

List of Figures

[Remove this section if it isn't needed —SS]

1 Symbols, Abbreviations, and Acronyms

symbol	description
T	Test

[symbols, abbreviations, or acronyms — you can simply reference the SRS (Cho et al., 2026b) tables, if appropriate —SS]

[Remove this section if it isn't needed —SS]

This document ... [provide an introductory blurb and roadmap of the Verification and Validation plan —SS]

2 General Information

2.1 Summary

FPVR is a virtual reality (VR)-based first-person-view (FPV) drone training simulator designed to help beginners develop and improve their drone piloting skills in a safe and accessible environment. Instead of requiring a physical drone, users can operate a simulated drone using VR controls and practise essential flight manoeuvres without the risks of equipment damage or unsafe flying conditions. It is also meant as a tool for experienced Drone fliers to be able to practice in a safe and controlled environment without the pressure of piloting a very expensive piece of equipment.

The system features free-flight practice, progressively challenging obstacle courses, multiple virtual environments, and checkpoints to create a structured learning experience. Users can track their performance through metrics such as completion time, checkpoint progress, and performance ratings, with potential AI-driven feedback to help identify areas for improvement. FPVR also supports registered accounts, saved performance records, attempt history, and local and global leaderboards, allowing users to monitor their progress and compare results with others. Overall, FPVR aims to make learning FPV drone piloting more accessible, engaging, and cost-effective through an immersive virtual training experience.

2.2 Objectives

[State what is intended to be accomplished. The objective will be around the qualities that are most important for your project. You might have something like: "build confidence in the software correctness," "demonstrate adequate usability." etc. You won't list all of the qualities, just those that are most important. —SS]

[You should also list the objectives that are out of scope. You don't have the resources to do everything, so what will you be leaving out. For instance, if you are not going to verify the quality of usability, state this. It is also worthwhile to justify why the objectives are left out. —SS]

[The objectives are important because they highlight that you are aware of limitations in your resources for verification and validation. You can't do everything, so what are you going to prioritize? As an example, if your system depends on an external library, you can explicitly state that you will assume that external library has already been verified by its implementation team. —SS]

2.3 Custom Documents

[Summarize the CDs (Custom Documents) that were tackled by this project. CDs can include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. CDs should have already been approved by the course instructor as included in your problem statement. You can use a pull request to update your CDs (in TeamComposition.csv or Repos.csv) if your plan changes as a result of the VnV planning exercise. —SS]

The project will include two Custom Documents: a Usability Report and a Design Thinking Report. The Usability Report will document a systematic and quantitative evaluation of FPVR's usability by having representative users complete predefined tasks within the simulator. The evaluation will establish testing goals, identify specific user tasks, collect feedback through structured questions and surveys, and measure usability outcomes to identify areas for improvement. The Design Thinking Report will document the human-centred design process used to understand the needs and challenges of beginner FPV drone pilots. This will include conducting user interviews, organizing findings through empathy maps, generating and evaluating potential solutions based on feasibility and desirability, and developing two rounds of prototypes to gather user feedback and refine the proposed design. These documents will help ensure that FPVR's design decisions are informed by user needs and that its usability is evaluated through structured user testing.

2.4 Relevant Documentation

This VnV plan is derived from two project documents: the Software Requirements Specification (SRS) (Cho et al., 2026b) and the Hazard Analysis (Cho et al., 2026a). Together they define what FPVR is supposed to do and which

failures would be unacceptable, which is the information needed to choose tests and interpret results.

The SRS (Cho et al., 2026b) is the statement of the software requirements. It records the goals of FPVR, the functional requirements (for example flight control, checkpoint behaviour, and session flow), and the non-functional requirements (for example performance, usability, and safety-related qualities). System tests in this plan are written against those requirements so that a passing test set is evidence that the implemented product matches the specified behaviour. The SRS also supplies identifiers, assumptions, and environmental constraints that this plan uses for traceability: each system test should map back to one or more requirements.

The Hazard Analysis (Cho et al., 2026a) identifies hazards and risks for FPVR. Through a Failure Mode and Effect Analysis it describes how components can fail, what the effects of those failures would be, and which safety and security requirements mitigate them. Those hazards and the derived safety requirements focus on the cases where incorrect behaviour could injure a user, damage equipment, produce misleading training, or compromise the integrity of the application. Tests inspired by the Hazard Analysis therefore include failure and mitigation scenarios (for example invalid controller input, lost tracking, or unrealistic physics) in addition to the happy-path cases taken from the SRS.

3 Plan

[Introduce this section. You can provide a roadmap of the sections to come. —SS]

3.1 Verification and Validation Team

[Your teammates. Maybe your supervisor. You should do more than list names. You should say what each person's role is for the project's verification. A table is a good way to summarize this information. —SS]

3.2 SRS Verification

[List any approaches you intend to use for SRS verification. This may include ad hoc feedback from reviewers, like your classmates (like your primary

reviewer), or you may plan for something more rigorous/systematic. —SS]

[If you have a supervisor for the project, you shouldn't just say they will read over the SRS. You should explain your structured approach to the review. Will you have a meeting? What will you present? What questions will you ask? Will you give them instructions for a task-based inspection? Will you use your issue tracker? —SS]

[Maybe create an SRS checklist? —SS]

3.3 Design Verification

[Plans for design verification —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.4 Verification and Validation Plan Verification

[The verification and validation plan is an artifact that should also be verified. Techniques for this include review and mutation testing. —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.5 Implementation Verification

[You should at least point to the tests listed in this document and the unit testing plan. —SS]

[In this section you would also give any details of any plans for static verification of the implementation. Potential techniques include code walkthroughs, code inspection, static analyzers, etc. —SS]

[The final class presentation in CAS 741 could be used as a code walkthrough. There is also a possibility of using the final presentation (in CAS741) for a partial usability survey. —SS]

3.6 Automated Testing and Verification Tools

[What tools are you using for automated testing. Likely a unit testing framework and maybe a profiling tool, like ValGrind. Other possible tools include a static analyzer, make, continuous integration tools, test coverage tools, etc.

Explain your plans for summarizing code coverage metrics. Linters are another important class of tools. For the programming language you select, you should look at the available linters. There may also be tools that verify that coding standards have been respected, like flake9 for Python. —SS]

[If you have already done this in the development plan, you can point to that document. —SS]

[The details of this section will likely evolve as you get closer to the implementation. —SS]

3.7 Software Validation

[If there is any external data that can be used for validation, you should point to it here. If there are no plans for validation, you should state that here. —SS]

[You might want to use review sessions with the stakeholder to check that the requirements document captures the right requirements. Maybe task based inspection? —SS]

[For those capstone teams with an external supervisor, the Rev 0 demo should be used as an opportunity to validate the requirements. You should plan on demonstrating your project to your supervisor shortly after the scheduled Rev 0 demo. The feedback from your supervisor will be very useful for improving your project. —SS]

[For teams without an external supervisor, user testing can serve the same purpose as a Rev 0 demo for the supervisor. —SS]

[This section might reference back to the SRS verification section. —SS]

4 System Tests

[There should be text between all headings, even if it is just a roadmap of the contents of the subsections. —SS]

4.1 Tests for Functional Requirements

[Subsets of the tests may be in related, so this section is divided into different areas. If there are no identifiable subsets for the tests, this level of document structure can be removed. —SS]

[Include a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. —SS]

4.1.1 Area of Testing1

[It would be nice to have a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. If a section covers tests for input constraints, you should reference the data constraints table in the SRS. —SS]

Title for Test

1. test-id1

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs. Output is not how you are going to return the results of the test. The output is the expected result. —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

2. test-id2

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

4.1.2 Area of Testing2

...

4.2 Tests for Nonfunctional Requirements

[The nonfunctional requirements for accuracy will likely just reference the appropriate functional tests from above. The test cases should mention reporting the relative error for these tests. Not all projects will necessarily have nonfunctional requirements related to accuracy. —SS]

[For some nonfunctional tests, you won't be setting a target threshold for passing the test, but rather describing the experiment you will do to measure the quality for different inputs. For instance, you could measure speed versus the problem size. The output of the test isn't pass/fail, but rather a summary table or graph. —SS]

[Tests related to usability could include conducting a usability test and survey. The survey will be in the Appendix. —SS]

[Static tests, review, inspections, and walkthroughs, will not follow the format for the tests given below. —SS]

[If you introduce static tests in your plan, you need to provide details. How will they be done? In cases like code (or document) walkthroughs, who will be involved? Be specific. —SS]

4.2.1 Area of Testing1

Title for Test

1. test-id1

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input/Condition:

   Output/Result:

   How test will be performed:

2. test-id2

Type: Functional, Dynamic, Manual, Static etc.

Initial State:

Input:

Output:

How test will be performed:

4.2.2 Area of Testing2

...

4.3 Traceability Between Test Cases and Requirements

[Provide a table that shows which test cases are supporting which requirements. —SS]

5 Unit Test Description

[This section should not be filled in until after the MIS (detailed design document) has been completed. —SS]

[Reference your MIS (detailed design document) and explain your overall philosophy for test case selection. —SS]

[To save space and time, it may be an option to provide less detail in this section. For the unit tests you can potentially layout your testing strategy here. That is, you can explain how tests will be selected for each module. For instance, your test building approach could be test cases for each access program, including one test for normal behaviour and as many tests as needed for edge cases. Rather than create the details of the input and output here, you could point to the unit testing code. For this to work, you code needs to be well-documented, with meaningful names for all of the tests. —SS]

5.1 Unit Testing Scope

[What modules are outside of the scope. If there are modules that are developed by someone else, then you would say here if you aren't planning on verifying them. There may also be modules that are part of your software, but have a lower priority for verification than others. If this is the case, explain your rationale for the ranking of module importance. —SS]

5.2 Tests for Functional Requirements

[Most of the verification will be through automated unit testing. If appropriate specific modules can be verified by a non-testing based technique. That can also be documented in this section. —SS]

5.2.1 Module 1

[Include a blurb here to explain why the subsections below cover the module. References to the MIS would be good. You will want tests from a black box perspective and from a white box perspective. Explain to the reader how the tests were selected. —SS]

1. test-id1

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

2. test-id2

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

3. ...

5.2.2 Module 2

...

5.3 Tests for Nonfunctional Requirements

[If there is a module that needs to be independently assessed for performance, those test cases can go here. In some projects, planning for nonfunctional tests of units will not be that relevant. —SS]

[These tests may involve collecting performance data from previously mentioned functional tests. —SS]

5.3.1 Module ?

1. test-id1

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input/Condition:

   Output/Result:

   How test will be performed:

2. test-id2

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input:

   Output:

   How test will be performed:

5.3.2 Module ?

...

5.4 Traceability Between Test Cases and Modules

[Provide evidence that all modules have been considered. —SS]

References

John Cho, Ibrahim Tariq, Vikram Chandar, Sankalpa Chhetri Dhakal, and Arjun Sasidaran. Hazard analysis for fpvr. https://github.com/SoftwareMandems/FPVR/blob/main/docs/HazardAnalysis/HazardAnalysis.pdf, 2026a.

John Cho, Ibrahim Tariq, Vikram Chandar, Sankalpa Chhetri Dhakal, and Arjun Sasidaran. Software requirements specification for fpvr. https://github.com/SoftwareMandems/FPVR/blob/main/docs/SRS/SRS.pdf, 2026b.

6 Appendix

This is where you can place additional information.

6.1 Symbolic Parameters

The definition of the test cases will call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance.

6.2 Usability Survey Questions?

[This is a section that would be appropriate for some projects. —SS]

7 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

7.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

7.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

7.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

8 Appendix — Reflection

[This section is not required for CAS 741 —SS]

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. What knowledge and skills will the team collectively need to acquire to successfully complete the verification and validation of your project? Examples of possible knowledge and skills include dynamic testing knowledge, static testing knowledge, specific tool usage, Valgrind etc. You should look to identify at least one item for each team member.

4. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?